

Hadoop Installation (Single Node Setup)

Step 1: Update Ubuntu

```
sudo apt update  
sudo apt upgrade -y
```

Step 2: Install Java

Check available Java version:

```
java -version
```

Install OpenJDK:

```
sudo apt install openjdk-11-jdk -y
```

Verify installation:

```
java -version
```

Find Java path:

```
readlink -f $(which java)
```

Example output:

```
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
```

JAVA_HOME will be:

```
/usr/lib/jvm/java-11-openjdk-amd64
```

Step 3: Create Hadoop User

```
sudo adduser hadoop
```

Give password and press Enter.

Add user to sudo group:

```
sudo usermod -aG sudo hadoop
```

Switch to Hadoop user:

```
su - hadoop
```

Step 4: Configure SSH

Install SSH:

```
sudo apt install openssh-server openssh-client -y
```

Generate SSH key:

```
ssh-keygen -t rsa -P ""
```

Press Enter for all prompts.

Enable passwordless SSH:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
chmod 600 ~/.ssh/authorized_keys
```

Test SSH:

```
ssh localhost
```

Type:

```
yes
```

Exit:

```
exit
```

Step 5: Download Hadoop

Move to home directory:

cd ~

Download Hadoop:

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
```

Extract Hadoop:

```
tar -xvzf hadoop-3.4.1.tar.gz
```

Rename folder:

```
mv hadoop-3.4.1 hadoop
```

Step 6: Configure Environment Variables

Open bashrc:

```
nano ~/.bashrc
```

Add these lines at the end:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=$HOME/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
```

Save and exit:

Ctrl + O

Enter

Ctrl + X

Apply changes:

```
source ~/.bashrc
```

Step 7: Configure hadoop-env.sh

Open file:

```
nano ~/hadoop/etc/hadoop/hadoop-env.sh
```

Find:

```
export JAVA_HOME=
```

Replace with:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Save and exit.

Step 8: Configure core-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/core-site.xml
```

Replace content between `<configuration>` tags:

```
<configuration>
```

```
<property>
```

```
<name>fs.defaultFS</name>
```

```
<value>hdfs://localhost:9000</value>
```

```
</property>
```

```
</configuration>
```

Save and exit.

Step 9: Configure hdfs-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/hdfs-site.xml
```

Replace content:

```
<configuration>

<property>
<name>dfs.replication</name>
<value>1</value>
</property>

<property>
<name>dfs.name.dir</name>
<value>file:///home/hadoop/hadoopdata/hdfs/namenode</value>
</property>

<property>
<name>dfs.data.dir</name>
<value>file:///home/hadoop/hadoopdata/hdfs/datanode</value>
</property>

</configuration>
```

Save and exit.

Step 10: Create HDFS Directories

```
mkdir -p ~/hadoopdata/hdfs/namenode
mkdir -p ~/hadoopdata/hdfs/datanode
```

Step 11: Configure mapred-site.xml

Copy template:

```
cp ~/hadoop/etc/hadoop/mapred-site.xml.template ~/hadoop/etc/hadoop/mapred-site.xml
```

Open file:

```
nano ~/hadoop/etc/hadoop/mapred-site.xml
```

Add:

```
<configuration>

<property>
```

```
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>

</configuration>
```

Save and exit.

Step 12: Configure yarn-site.xml

Open file:

```
nano ~/hadoop/etc/hadoop/yarn-site.xml
```

Add:

```
<configuration>

<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>

</configuration>
```

Save and exit.

Step 13: Format NameNode

```
hdfs namenode -format
```

Successful output:

Storage directory has been successfully formatted

Step 14: Start Hadoop Services

Start HDFS:

start-dfs.sh

Start YARN:

start-yarn.sh

Step 15: Verify Running Services

jps

Step 16: Check Hadoop Version

hadoop version

Step 17: Access Hadoop Web Interfaces

NameNode UI:

<http://localhost:9870>

ResourceManager UI:

<http://localhost:8088>

Step 18: Stop Hadoop

Stop YARN:

stop-yarn.sh

Stop HDFS:

stop-dfs.sh

Quick Verification Commands

```
vikas@Vikash:/mnt/c/Users/Vikas$ jps
7107 SecondaryNameNode
6739 NameNode
7463 NodeManager
6871 DataNode
11688 Jps
7338 ResourceManager
vikas@Vikash:/mnt/c/Users/Vikas$ |
```